

Graphs, Colouring & Incidence Counting

Model the relation first; count only after the graph is correct

OBJECTS → RELATIONS → GRAPH → INVARIANT → CHECK

Student learning pack · assimilation, recognition, transfer, mastery

Graphs, Colouring and Incidence Counting: Model the Relation First

A graph is not mainly a picture of dots and lines. It is a way to keep exactly the relationships that matter and throw away the rest.

The central habit is:

Before using graph vocabulary, decide what one vertex means and what one edge means.

A useful working loop is:

DEFINE OBJECTS -> MODEL RELATIONS -> CHOOSE DEGREE / COLOUR / INCIDENCE VIEW -> COUNT OR FORCE -> CHECK DOUBLE COUNT

1. When a graph is the right representation

Suppose a problem is about five people and which pairs know each other. Their heights, seating positions, and names may be irrelevant. If only pairwise acquaintance matters, represent:

- one person by one **vertex**;
- one acquaintance relation by one **edge**.

Now the surface story has become a relation structure.

A graph is useful only if the model preserves every condition needed for the question.

2. Vertices, edges and adjacency

Two vertices are **adjacent** when an edge joins them.

The **degree** of a vertex is the number of edges incident with it.

If a vertex has degree 4, it is related to four other vertices under the chosen model.

Do not count degree from the visual length or direction of an edge. Only incidence matters.

3. The handshaking identity

Every edge has two endpoints. Therefore, when we add all vertex degrees, each edge is counted twice:

sum of all degrees = 2 x number of edges.

This is the handshaking identity.

Example

A graph has degrees

3, 3, 2, 2, 2, 2.

Their sum is 14, so the graph has

$14/2 = 7$

edges.

The factor 2 is not a formula to memorize blindly. It comes from counting each edge once at each endpoint.

4. Direct edge count or degree sum?

Use direct enumeration when edges fall naturally into a few disjoint types.

Use degree sum when local degrees are easier to compute than the global edge set.

Ask:

Is it easier to count relations themselves, or count how many relations touch each object?

5. Proper colouring is not unrestricted colouring

In a vertex-colouring problem, colours are assigned to vertices.

A **proper colouring** requires adjacent vertices to have different colours.

If a graph has 4 vertices and 4 available colours, there are 4^4 unrestricted assignments. A proper colouring is a smaller set because adjacency forbids some assignments.

The first step is not “use a colouring formula.” It is:

1. identify the graph;
2. identify adjacency;
3. count colour choices under those restrictions.

6. Sequential colouring: restrictions change after each choice

Consider a triangle with 4 colours.

- first vertex: 4 choices;
- second adjacent vertex: 3 choices;
- third vertex adjacent to both: 2 choices.

So there are

$$4 \cdot 3 \cdot 2 = 24$$

proper colourings.

This uses the ordinary multiplication principle already learned in counting. The new content is the graph restriction that determines the number of legal choices at each stage.

7. A missing edge changes the colouring count

Suppose four vertices form a quadrilateral and one diagonal is present. This is K_4 with one edge missing.

The two non-adjacent vertices are allowed to share a colour. That single missing relation matters.

In IOQM-2025-Q08, modelling the picture this way turns the problem into a short proper-colouring count. The answer is 48.

The lesson is:

Colour the graph you actually have, not the complete graph your eye imagines.

8. Cyclic colouring: the end must meet the beginning

A line of positions and a cycle of positions can have the same local-looking rule but different global constraints.

For a cyclic arrangement, positions near the end may also be close to positions near the beginning.

If every set of five consecutive vertices must have different colours, then vertices within cyclic distance 4 must receive different colours. This is naturally a graph-colouring condition.

The wrap-around condition is part of the problem, not a final correction.

9. Grid and knight graphs

A board problem can become a graph:

- each square is a vertex;
- two squares are adjacent if a legal knight move connects them.

Then “how many pairs of squares are connected by a knight move?” becomes “how many edges are in the knight graph?”

You may count directed moves from each starting square and divide by 2, but only because every unordered legal pair is counted once from each endpoint.

In IOQM-2024-Q09, this degree/incidence viewpoint gives 48 unordered knight-move pairs on the 5 x 5 board.

10. Ordered moves vs unordered pairs

If you count

A $\rightarrow$ B

and

B $\rightarrow$ A

as two directed moves, then an undirected edge $\{A, B\}$ has been counted twice.

But you must not divide by 2 automatically. Ask:

Does every final object appear exactly twice in my count?

This is the same counting discipline as before, now expressed through graph incidence.

11. Incidence double counting

An **incidence** is a relation such as:

- vertex lies on an edge;
- point lies on a line;
- chosen cevian passes through an intersection;
- region boundary contains an edge.

Double counting means counting the same set of incidences in two different ways.

The most important first step is to name the incidence object clearly.

Example

If a graph has E edges, then the set

$\{(\text{vertex}, \text{edge}): \text{vertex is an endpoint of edge}\}$

has size $2E$ when counted by edges, and sum degrees when counted by vertices. This is exactly the handshaking identity.

12. Geometry can hide an incidence problem

A problem may mention points on triangle sides, cevians and regions. The geometry may be only the surface.

If the question depends on which cevians cross, concur, or create new regions, an incidence representation may be cheaper than coordinates.

In IOQM-2023-Q22, the requirement of exactly nine regions can be translated into the required intersection pattern. Once that pattern is understood, the remaining task is a count of valid peg selections. The answer is 77.

The lesson is:

When the exact lengths and angles do not matter, ask whether only the intersection pattern matters.

13. Edge colouring and monochromatic triangles

Sometimes colours belong to edges rather than vertices.

A triangle is **monochromatic** when all three of its edges have the same colour.

A raw search over all edge colourings is usually expensive. Instead, fix one vertex and look at the colours of the edges incident with it.

If many incident edges share one colour, the edges among their other endpoints are forced to avoid that colour if monochromatic triangles are forbidden.

This is a local-to-global forcing argument.

14. A Ramsey-style inevitability habit

Ramsey-style reasoning does not mean memorizing large Ramsey numbers. At this level, it means:

1. fix a small local structure;
2. use pigeonhole or degree information to force a repeated type;
3. examine the relations among the forced neighbours;
4. derive either the required configuration or a contradiction.

In IOQM-2024-Q19, forbidding monochromatic triangles in a red/blue colouring of the edges of K_5 forces a very rigid cycle-like pattern. The answer is 12.

The lesson is:

Do not enumerate every colouring when local avoidance rules force the rest.

15. Forbidden subgraphs

A colouring condition can often be restated as “a certain small graph must not appear in one colour.” For example, “every triangle has at least one red edge” means the blue-edge graph contains no triangle.

In IOQM-2023-Q16, the sides of the hexagon are already red, and the diagonal colours must avoid creating an all-blue triangle. The graph language makes the forbidden configuration explicit. The answer is 94.

16. Cyclic local restrictions as graph powers

Suppose vertices of a regular polygon are coloured and any five consecutive vertices must all have different colours.

Two vertices are forbidden to share a colour whenever their cyclic distance is at most 4.

Instead of repeatedly speaking about overlapping blocks of five, build a graph in which those forbidden pairs are adjacent. Then the problem is an ordinary proper-colouring question on that graph.

This is the key representation in IOQM-2025-Q29, whose answer is 19.

The lesson is:

A repeated local restriction can become one static adjacency rule.

17. Graph state vs game state

A graph can represent legal moves in a game, but a static graph-colouring/counting question is not automatically a game-strategy problem.

If players make choices adversarially and future options depend on the history, the central object is a game state and winning/losing strategy. That belongs to a different route.

If the task only asks which assignments or relations are valid in a fixed graph, stay in graph counting/colouring.

18. Historical pattern: knight moves

IOQM-2024-Q09 rewards a degree/incidence count rather than listing every pair individually.

The lesson is:

When local move counts are easy, sum local degrees and correct the double count.

19. Historical pattern: proper colouring

IOQM-2025-Q08 looks like a small colouring exercise, but the important step is identifying the exact adjacency graph before multiplying choices.

The lesson is:

Restrictions come from edges, not from the number of colours alone.

20. A compact router

When a relation/counting problem appears, ask:

1. What should one vertex represent?
2. What exactly should an edge represent?
3. Does the graph preserve every relevant condition?
4. Is the target an edge count, degree count, colouring, or incidence count?
5. If colouring, is it proper and is the arrangement cyclic?
6. If counting moves, am I counting directed moves or unordered pairs?
7. Can the same incidence set be counted in two ways?
8. Is a forbidden local configuration forcing a global pattern?
9. Is this actually an adversarial game rather than a static graph problem?

Build the right representation first. The counting becomes shorter afterward.

First-Step Reference: Graphs, Colouring and Incidence Counting

Use the smallest representation that preserves the restriction. Do not introduce graph terminology unless it simplifies the problem.

You see	First question	First mathematical line
objects with pairwise relations	what should a vertex and edge mean?	write <i>vertices</i> = ..., <i>edges</i> = ...
graph degrees	are local degrees easier than listing edges?	write $\text{sum degrees} = 2 E $
coloured vertices	are adjacent vertices required to differ?	write the adjacency restriction before counting choices
a nearly complete small graph	which edges are missing?	identify the exact non-adjacent pairs
positions around a polygon/cycle	does the restriction wrap around?	translate cyclic distance into adjacency
legal moves on a board	am I counting directed moves or unordered pairs?	define the move graph and what one edge means
directed move total used for unordered pairs	is every edge counted from both ends exactly once?	divide by 2 only after verifying the two-count
regions/intersections/cevians	do exact lengths matter, or only who meets whom?	define one incidence/intersection object
same incidence set can be counted locally and globally	what are the two viewpoints?	write both counts for the same set before equating
red/blue edge colouring with forbidden monochromatic triangle	what happens at one fixed vertex?	group incident edges by colour
every small local block must use distinct colours	can forbidden pairs be made edges of one graph?	build the static conflict graph
players choose moves adversarially	is this really a fixed colouring/count?	route to game-state analysis instead of static graph counting

Fast route choices

Direct enumeration vs degree sum

Use **direct enumeration** when edges split naturally into a few disjoint types.

Use **degree sum** when the number of legal neighbours of each vertex or vertex class is easy to compute.

Unrestricted assignments vs proper colouring

Start with unrestricted assignments only as a comparison. The actual count must respect every adjacency restriction.

Linear vs cyclic colouring

On a line, the last position may be unrelated to the first.

On a cycle, wrap-around constraints are active from the start. Do not “fix the ends” after finishing a linear count unless you have proved that correction is valid.

Static graph vs game state

A fixed graph records allowed/forbidden relations.

A game state records whose turn it is, what moves remain, and how choices affect the future. If adversarial strategy matters, a static graph count is not enough.

The three checks before dividing by 2

Before halving a count, confirm:

1. every desired unordered object appears in the directed/incidence count;
2. every desired object appears exactly twice;
3. no loop or exceptional object is counted a different number of times.

The three checks before a colouring product

Before multiplying available colour choices, confirm:

1. the graph and adjacency are correct;
2. each stage count reflects all colours already forbidden by coloured neighbours;
3. cyclic closure or other global restrictions have not been postponed incorrectly.

Stop rule

Stop once the requested count or forced structure is determined and the counted object has been checked for overcount/undercount. More graph vocabulary is not more proof.

Recognition and First-Line Lab

For each item, write only the **best first move** or choose the best option. Do not complete a long solution unless needed to justify the choice.

Recognition

1. A problem gives people and which pairs exchange messages. Which graph model is natural? A. vertices are messages B. vertices are people and edges are communicating pairs C. edges are people D. colours are messages
2. You know every vertex degree but not the edge list. Which identity should you test first? A. inclusion-exclusion B. $\sum \text{degrees} = 2|E|$ C. Pythagoras D. a recurrence
3. Four colours are assigned to vertices, and adjacent vertices must differ. What is being counted? A. unrestricted assignments B. proper vertex colourings C. edge lengths D. game states
4. A graph is “complete except for one missing edge.” Why does the missing edge matter in a colouring count? A. its endpoints may be allowed to share a colour B. it changes the number of colours C. it creates a loop D. it forces every colour to be used
5. A local colour restriction is stated around a polygon. What must be checked before treating it like a line of positions? A. whether colours are integers B. cyclic wrap-around C. vertex coordinates D. polygon area
6. You count every legal knight move from every starting square, but the question asks for unordered pairs of squares joined by a knight move. What must be verified before halving? A. every pair was counted exactly twice B. the board is square C. the knight is centred D. every square has equal degree
7. A geometry problem asks about cevians and regions, but lengths and angles never enter the answer. Which representation may be cheaper? A. incidence/intersection graph B. trigonometry C. coordinates of every point D. mensuration
8. Players alternately choose legal moves and try to force a win. Which warning is correct? A. every game is just a proper colouring B. a static graph count may not capture turn/history information C. degree sum always gives the winner D. ignore strategy

First lines

9. Six students are connected when they are friends. Write one sentence defining the vertices and edges.
10. A graph has degrees 4, 3, 3, 2, 2, 2. Write the line that determines the number of edges.
11. A triangle is properly coloured with 5 available colours. Write the stage-by-stage choice product, without evaluating it.
12. Vertices of a cycle must have different colours whenever their cyclic distance is at most 2. State which pairs become edges in the conflict graph.
13. A board problem asks for unordered knight-move pairs. Write the graph interpretation of one vertex and one edge.
14. Eight lines each pass through three selected points, and every selected point lies on exactly two of the lines. Define the incidence set you would count two ways.
15. The edges of K_5 are red/blue and monochromatic triangles are forbidden. State the local structure you should inspect first.
16. A token-moving problem is adversarial. Write the extra information a game state needs beyond the static move graph.

Practice and Transfer Bank

Try each stage without looking ahead. The prompts become less explicit as you move down the page.

Reconnect

1. A graph has vertex degrees 2, 2, 3, 3. How many edges does it have?
2. Can a graph have degree sequence 3, 3, 3, 2? Give a one-line reason.
3. A path with four labelled vertices is properly coloured with three available colours. How many colourings are possible?
4. A triangle is properly coloured with four available colours. How many colourings are possible?

Structure with support

5. A graph is K_4 with exactly one edge removed. Find the number of proper colourings using three colours.
6. How many edges are in $K_{3,3}$? Verify your result using degrees.
7. How many unordered pairs of squares on a 4×4 board are connected by a knight move?
8. How many edges are in the complete bipartite graph $K_{3,4}$? Verify by handshaking.

Reduced support

9. Find the number of proper colourings of a 5-cycle using three colours.
10. A graph has degrees 1, 2, 2, 3, 4. Determine the number of edges.
11. Eight lines each contain exactly three selected points, and every selected point lies on exactly two of the lines. How many selected points are there?
12. How many unordered knight-move pairs are there on a 5×5 board?

Mixed independent

13. Prove that a finite graph cannot have exactly one vertex of odd degree.
14. Twelve labelled positions lie on a cycle. Colour them using exactly three named colours so that every three consecutive positions have pairwise different colours. How many valid colourings are possible?
15. Prove that every red/blue colouring of the edges of K_6 contains a monochromatic triangle.
16. Ten clubs each have exactly six student memberships counted with repetition across clubs. Every student belongs to exactly three clubs. How many distinct students are represented by these memberships?

Transfer

17. Historical anchor — IOQM-2025-Q08.

A quadrilateral has all four boundary edges and one diagonal. Its four vertices are coloured using four colours, and endpoints of every listed edge must receive different colours. Determine the number of valid colourings.

18. Historical anchor — IOQM-2024-Q09.

Determine the number of unordered pairs of squares on a 5×5 board that are connected by a knight move.

19. Historical anchor — IOQM-2024-Q19.

The edges of K_5 are coloured red or blue with no monochromatic triangle. Determine the number of valid colourings.

20. Historical anchor — IOQM-2025-Q29.

A regular polygon is called colourful if its vertices can be coloured with at most six colours so that any five consecutive vertices have pairwise distinct colours. Determine the largest number of vertices for which the polygon is not colourful.

21. Author-created forbidden-subgraph transfer.

A hexagon has all six boundary edges fixed red. Its nine diagonals may be red or blue. Count the diagonal colourings for which every triangle determined by three vertices has at least one red side.

22. Author-created incidence transfer.

Four segments are drawn inside a triangle, each from a vertex to a point on the opposite side. Suppose two segments start at vertex A and two at vertex B, with all four side-points distinct and no three segments concurrent. How many regions are formed? Justify using the intersection pattern rather than a drawing.

Historical geometric-surface anchors that require exact page/figure custody remain in the source map and are not promoted verbatim here until that custody gate closes.

Benchmark Assimilation Lab — Graphs, Colouring and Incidence Counting

Use this after the Assimilation Book and before the final mastery test. The aim is to test whether you can build the correct model, not whether you remember graph vocabulary.

Learning loop:

RECONNECT -> DIAGNOSE -> ADOPT -> TRANSFER

A. RECONNECT — no notes

Write only the first useful line unless a short answer is requested.

1. In a friendship problem, what must be defined before writing a graph?
2. Why is the sum of all vertex degrees even?
3. What is the difference between unrestricted vertex colourings and proper colourings?
4. Why can a linear colouring count fail on a cycle?
5. Before dividing a directed-move count by 2, what must be proved?
6. If a geometry problem depends only on crossings and concurrencies, what representation should be tested before coordinates?
7. In a red/blue edge-colouring problem forbidding monochromatic triangles, what local structure should you inspect first?
8. What extra information does a game state need beyond the underlying move graph?

Diagnostic interpretation:

- misses 1 -> graph-modelling bridge is weak;
- misses 2 or 5 -> incidence/double-count bridge is weak;
- misses 3-4 -> colouring restriction/closure bridge is weak;
- misses 6 -> representation-selection bridge is weak;
- misses 7 -> Ramsey local-forcing bridge is weak;
- misses 8 -> static-graph vs game-state boundary is weak.

B. Error laboratory

For each proposed solution, identify the first invalid or inefficient move and repair it.

Error 1 — drawing before defining

Claim: “The story has six people, so draw six dots in a hexagon and connect the obvious pairs.”

Repair target: define exactly what one vertex and one edge mean before drawing.

Error 2 — unrestricted colouring reflex

Claim: “Four vertices, three colours, so there are 3^4 colourings.”

Repair target: determine whether adjacency restrictions make the colouring proper.

Error 3 — divide by two automatically

Claim: “I counted 40 moves, so there are 20 edges.”

Repair target: prove every desired unordered edge was counted exactly twice and no exceptional object behaves differently.

Error 4 — linearize a cycle

Claim: “Colour the first vertex, then continue left to right; the last position has no extra condition.”

Repair target: include wrap-around adjacency from the beginning.

Error 5 — count the wrong incidence set

Claim: “I counted point-line incidences by lines and counted crossing points by points, so the two totals are equal.”

Repair target: explicitly define one identical incidence set before double counting.

Error 6 — brute-force Ramsey search

Claim: “There are only 2^{10} colourings of the edges of K_5 , so enumerate all of them.”

Repair target: fix one vertex and force the local colour pattern first.

C. ADOPT — first move only

Do not solve fully. Write the first two useful lines.

1. Six students exchange messages; only who communicates with whom matters.
2. A graph has degree sequence 4, 4, 3, 3, 2, 2.
3. Colour the vertices of a path of length 5 properly with 4 colours.
4. Colour the vertices of a 7-cycle properly with 3 colours.
5. Count unordered knight-move pairs on a 6×5 board.
6. Ten lines each contain 4 selected points; each selected point lies on exactly 2 lines.
7. Two colours are used on the edges of K_6 ; prove a monochromatic triangle must occur.
8. A polygon rule says any four consecutive vertices must have distinct colours. What static conflict graph should be built?

D. TRANSFER — changed surface

1. A tournament schedule says each team plays certain pairs of opponents exactly once. The question asks whether a claimed list of numbers of games played is possible. Translate the claim into degree language before checking it.
2. A seating rule forbids guests within cyclic distance 2 from sharing a badge colour. Replace the repeated local rule by one graph-colouring statement.
3. A board has several legal jump types. Explain two different valid edge-count routes and the exact condition under which their totals must agree.
4. A family of sets is described by which elements belong to which sets. Define an incidence set that can be counted by sets and by elements.
5. A geometric diagram contains many cevians, but all lengths and angles are irrelevant to the target region count. Explain which information the graph/incidence model must preserve and which it may discard.
6. Construct a static graph-counting problem that looks game-like but does **not** require game-state analysis, then explain the boundary.

E. Six-question assimilation test

Choose one problem from Section D and answer all six questions before completing the count/proof.

1. **What did you notice?** State the relation pattern that matters.
2. **Why does the representation work?** Explain what information it preserves.
3. **What clue would make you think of it again?** Give a future recognition cue.
4. **What similar-looking situation needs a different method?** State one contrast pair.
5. **Can you write the first two useful lines without help?** Write them.
6. **Can you solve a disguised version?** Change the surface while preserving the same graph/incidence invariant and outline the route.

Readiness rule

You are not ready merely because you know the words vertex, degree, colouring or Ramsey. You are ready when you can define the relation model from an unfamiliar story, choose the smallest graph/incidence view, justify every overcount correction, handle cyclic closure, reject a tempting wrong representation, and start a disguised problem without a chapter label.

Mixed Mastery Test

Solve each problem without method labels or hints. Define the counted object or graph clearly whenever the representation is not already explicit.

1. A graph has degrees $3, 3, 2, 2, 2, 2$. Determine the number of edges.
2. Decide whether a graph can have degree sequence $4, 4, 3, 3, 3, 2$. Justify.
3. A path with five labelled vertices is properly coloured using four available colours. Determine the number of colourings.
4. Determine the number of proper colourings of a 4-cycle using three available colours.
5. A graph is K_4 with exactly one edge removed. Determine the number of proper colourings using four colours.
6. Determine the number of unordered knight-move pairs on a 4×4 board.
7. Ten clubs each contribute six student-membership incidences. Every student belongs to exactly three clubs. Determine the number of students.
8. Eight labelled positions lie on a cycle and three named colours are available. Every three consecutive positions must have pairwise distinct colours. Determine whether a valid colouring exists and justify your conclusion.
9. Prove that every red/blue colouring of the edges of K_6 contains a monochromatic triangle.
10. Twelve lines each contain exactly four selected points. Every selected point lies on exactly three of the lines. Determine the number of selected points.
11. A directed graph records all legal moves of a two-player game. Explain why counting outgoing edges alone is not enough to determine a winning strategy, and name at least two additional pieces of state information that may matter.
12. Determine the number of proper colourings of a 7-cycle using three available colours.